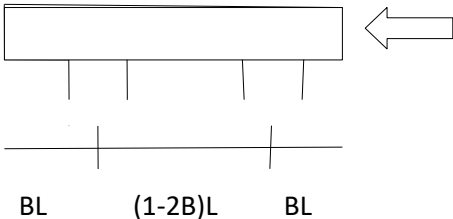


Hydraulics

1. When we throw a piece of stone in the LakeTana it will sink down. But big boat like Tananesh floats in this lake. Describe briefly the phenomena happen here.

Ans:- Buoyancy's effect

Structure

2.  An overhang beam is shown in fig. have a dead load of 20kn/m and imposed load of 12kn.

A. calculate the value of B in which the support moment and the mid span moment is equal.

B. draw the IL at the mid span for shear and BM at support quantitatively

C. quantitatively draw the design bending moment diagram and SFD

D. what is the problem of having along cantilever (adv.&disadv)

Highway

3. as a civil engineering graduate

a. write at least two methods for setting out of a horizontal curve

ans: - theodolite method(angle& radius method)and tapping

b. what is the reason that surface dressed pavement is preferable than a bitumen asphalt pavement.

Ans:- cost, frequently "open" in character and has excellent nonskid and visibility characteristics, water proof characteristics, the road surface arrests deterioration, its strong enough to resist traffic forces and remain flexible at the lowest temperature.

HS

4. You are put in position where an earthen dam gone be built as a dam engineer give a brief answer for the following

a. what are the important data that you should have&what is the use of each data.

Ans:- topography, sediment rate, flood routing, spillway site, availability of construction materials, environmental considerations, submergence area, water tightness of reservoir, suitable foundation, overall cost.

b.what is the importance of the core

ans:- the core keeps the phreatic line within the dam and it helps to check the seepage.

C.what is the importance and position of filter materials

ans:- properly designed graded filters are required between the drain system and the adjoining soil to prevent the migration of soil particles into the drains which may cause piping. Filters are provided between the core of fine grained soils and the shells of the coarse-grained soil to prevent migration of particles from impervious zone to the pervious zone when the seepage flow takes place.

d.what is the use and positioning of spillway

ans:- the spillway is a hydraulic structure intended to discharge the excess water from the reservoir, during the time of flood, in such manner so as to insure the safety of the dam.

=> For earth dam outside the dam separate spillway must be constructed to avoid over flow dam. And for concrete dam crest spillway is used since it can stand overtopping.

e.in what way can we calculate the dam capacity

ans: - from flood routing, inflow flood hydrograph, reservoir capacity curve and outflow discharge curve.

Material+highway

5.as a material engineer

a.what tests should be performed on the concrete before mix design is applied on the field.

=> Sieve analysis in aggregates

b.what are the tests performed on the cement.

=> Normal consistency and setting times of concrete, fineness of cement, soundness test of cement, strength of cement.

RC

6. The consultant engineer design a '5' story building. The functional importance of all floors is the same residence. But the designer gives more thickness for first floor slab. As a civil engineer you take position here to supervise that project. Will you approve that design or not?

Ans:- the slab must not have different thickness.

RC

7. A building having a basement faces a problem of damping in the basement floor side wall. Without damping the structural system of the building;what you recommend to overcome this problem.

Ans: - piping the water out of the damp area, additional water proofing materials can be used to cover the retaining water (eg. Laminate it by grease, bituminous; cement routing, cover with PVC)

SOIL

8. a two story residential building overlaying on a 6m sand and gravel mixture. The water table is 4m below NGL and 2m clay below the sand gravel soil. A building exerts 200kpa on the middle of the clay. Adjacent to this building a basement for parking is going to be built at 3m height retaining structure is built to support side flow of soil.

a. what are the important soil parameters needed for design and their use

ans:- cohesion and internal friction maybe unit weight

b. what are the important soil tests that should be done design input and describe briefly each test

ans:- compaction test, optimum moisture content, specific gravity, plasticity limit and liquid limit, penetration test.

c. what will happen if we use the cohesive as back fill

material

ans:- cohesive soil will form tension crack on the surface and it won't resist the pressure applied in to it. And it must be non-cohesive since it have better strength and can resist the pressure.

HIGHWAY

9. As we pass through the high way from A.A to B/dr we find AbayBerha which is now one of the most place in which a high way engineers and geologist are studied very carefully about the fail of the slope and the like. As a civil engineer you are in position to comment your idea to protect the road

a. what should be done to prevent the soil stability problem

ans:- adding drainage system using pipes

b. what is the reason that the failure is common in summer time

ans:- ground

water fluctuation due to rain.

c. what type of slope failure is it

ans:- rotational failure

HS

10. The bridge which connects Abaymado with the rest of B/d constructed on Abay river has one span with 7m width and 6m length 200mm thickness. Calculate the dead load per unit width if the density of the concrete is 2400 kg/m^3 .

Ans:-